Windows Brute-Force Authentication Investigation

Portfolio case study | Wazuh SIEM + Windows 11 | October 16-17, 2026

Case ID	SOC-001
Disposition	Authorized simulation; detection validated
Wazuh severity	Level 10 (rule 100004)
Affected host	SOC-WIN11 / agent 001 / 192.168.x.x
Target account	SOC-WIN11\WazuhTest
MITRE ATT&CK	T1110 - Brute Force

Executive summary

A controlled PowerShell test generated six failed Windows authentication attempts against the local account WazuhTest. Windows recorded Event ID 4625, and Wazuh correlated five failures within 60 seconds to trigger custom Level 10 rule 100004. The alert fired before the local Windows account-lockout threshold of ten attempts was reached.

A follow-up review of Wazuh alerts and the original Windows Security log found no matching successful logons (4624) or account lockouts (4740) in the reviewed time window. The test originated locally from IPv6 loopback ::1. This was an authorized simulation, not evidence of an external intrusion.

Scope and evidence

Investigated the SOC-WIN11 endpoint and Wazuh Manager alert records for the October 16 local-time / October 17 UTC test. Evidence includes the full custom-rule alert JSON, an alert timeline query, endpoint Security log queries, local account status, and the net accounts policy output. Screenshots are reproduced in the appendix.

Detection logic

Custom Wazuh rule 100004 correlates built-in authentication-failure rule 60122 using frequency=5, timeframe=60, and the same win.eventdata.targetUserName. It is mapped to MITRE ATT&CK; T1110. Detection alerts on suspicious frequency; it does not block sign-in attempts.

EVENT RECONSTRUCTION

Validated Wazuh alert timeline

Wazuh timestamp (UTC)	Rule	Observation
01:18:36.431	60122	Failed logon 1
01:18:39.455	60122	Failed logon 2
01:18:42.473	60122	Failed logon 3
01:18:45.500	60122	Failed logon 4
01:18:48.503	100004	Fifth failure; correlation alert
01:18:51.520	60122	Failed logon 6

The fifth event appears as the custom correlation alert rather than a separate rule 60122 entry. The rule 100004 alert contains the fifth Event ID 4625 as its full log and references earlier failures in previous_output.

Key authentication artifacts

Field	Observed value / interpretation
Windows Event ID	4625 - failed account logon
Target / subject	WazuhTest / labuser
Source address	::1 - IPv6 loopback (local origin)
Logon type	2 - interactive
Logon process	seclogo
Authentication package	Negotiate
Status / substatus	0xC000006D / 0xC000006A - authentication failure / incorrect password
Process reported	C:\Windows\System32\svchost.exe

Authentication and lockout validation

A query of Wazuh alerts.json found no matching 4624 alerts for WazuhTest from 01:18 to 01:30 UTC. An endpoint Get-WinEvent query found no matching 4624 Security events from 18:18 to 18:30 in the Windows local-time view. A separate endpoint query found no matching 4740 lockout events in that window.

The Get-LocalUser query confirmed the account existed, was enabled, required a password, and showed a blank LastLogon field. Account enabled status does not independently establish lockout state.

FINDINGS & RECOMMENDATIONS

Windows local account policy

Policy setting	Observed value
Lockout threshold	10 invalid attempts
Lockout duration	10 minutes
Lockout observation window	10 minutes
Minimum password length	0
Password history	None
Maximum password age	42 days

The six simulated failures were below the configured threshold of ten, consistent with the absence of a lockout event. The threshold was read after the test; the evidence does not independently prove the policy was identical at the time of the attempts.

Analyst assessment

Classification: Authorized lab simulation / expected positive detection. Detection outcome: Successful. Observed impact: No matching successful authentication or account lockout found in the specified review window. Compromise assessment: No evidence of successful compromise identified in the reviewed telemetry; this is not proof of absence outside the scope or time window.

Recommended improvements

• Keep the correlation rule and test it periodically with documented benign simulations.

• Correlate 4625 failures with 4624 successes for the same account, host, and time window.

• Review lockout threshold, duration, and observation window against operational needs; avoid overly aggressive settings that create denial-of-service risk.

• Review password policy: a minimum length of zero and no password-history requirement are weak baseline settings.

• Preserve raw Windows Security events and normalized SIEM alerts; capture host time-zone settings when producing incident timelines.

• Expand future tests to remote logons, multiple targeted accounts, and false-positive handling.

Evidence integrity and limitations

The alert timestamp and event systemTime are in UTC; screenshots of Windows queries used the endpoint local-time view. Time-zone settings were not independently captured in the evidence. The Wazuh alert file is filtered and does not necessarily retain every Windows Security event. A controlled six-attempt script was used, so the result demonstrates detection engineering and investigation rather than real adversary attribution.

Final disposition

Closed as authorized simulation. Rule 100004 detected the intended authentication-failure pattern. No successful logon was identified in the checked interval. Further hardening recommendations are recorded for the lab.

EVIDENCE APPENDIX

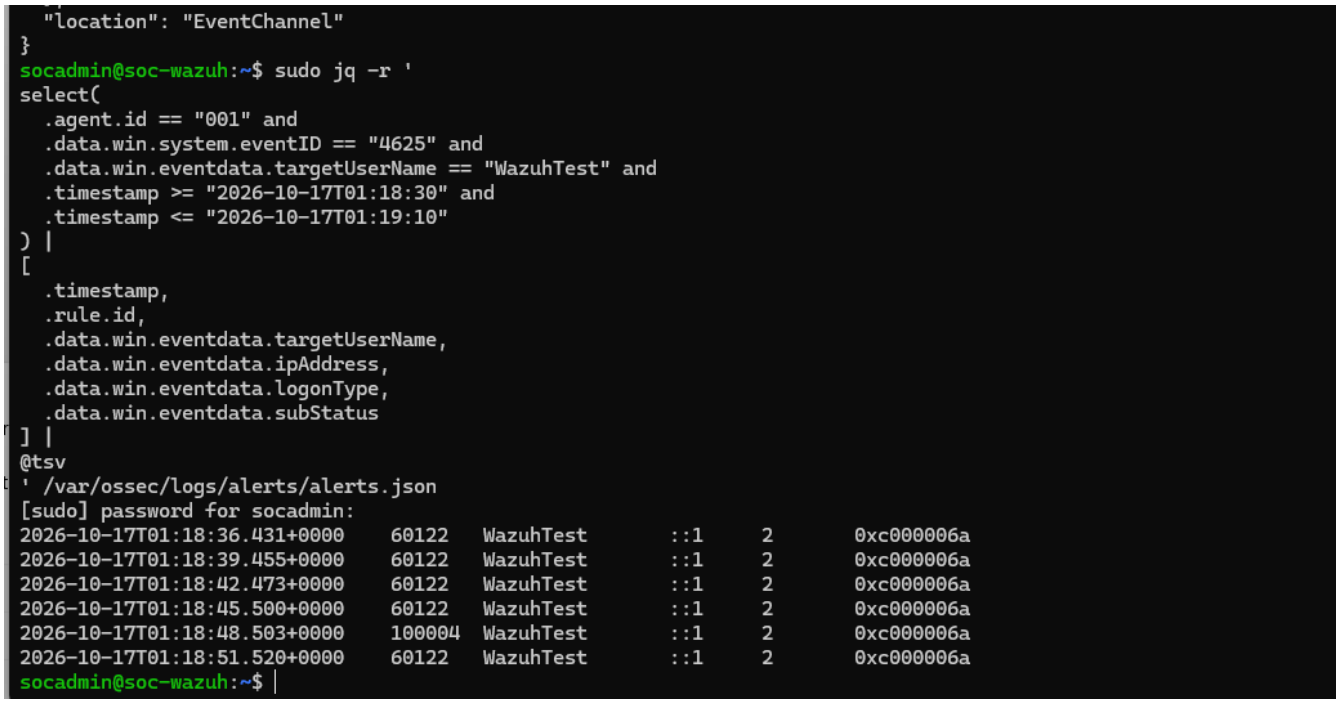


Exhibit A - Wazuh alerts.json timeline query: six failures with rule 100004 on the fifth event.

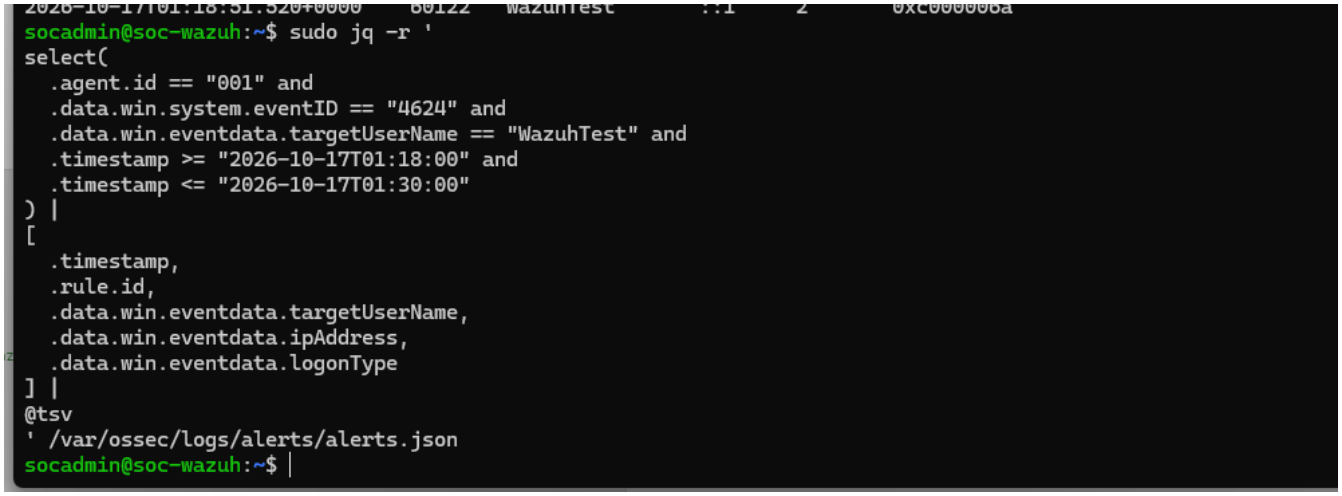


Exhibit B - Wazuh query for matching successful logon alerts (4624): no output.

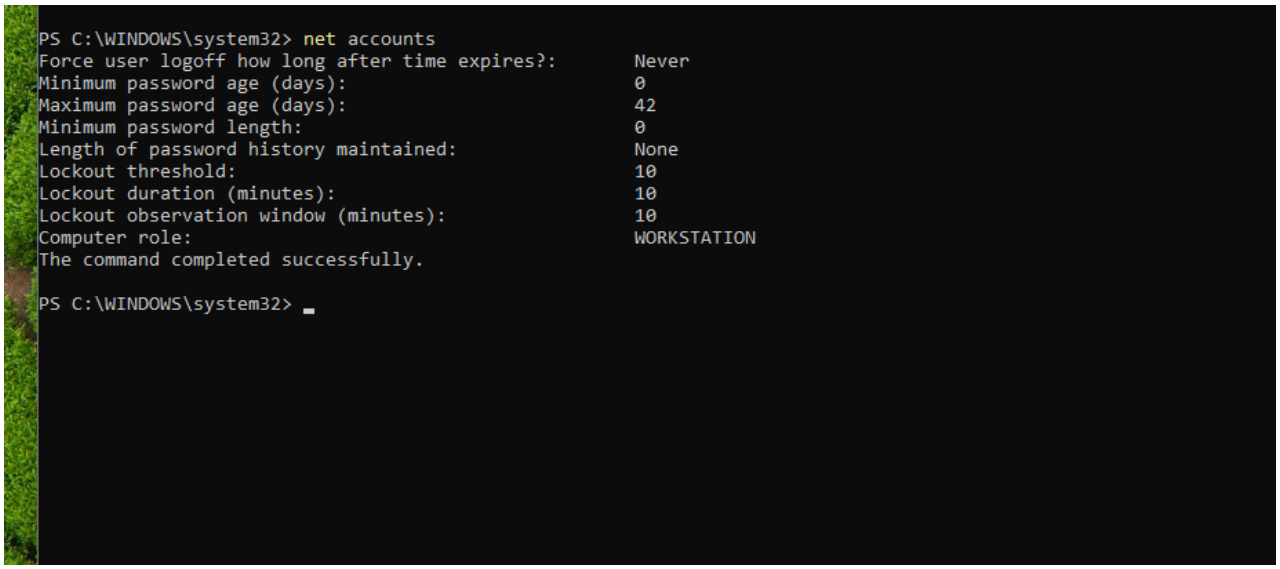


Exhibit C - Windows net accounts: 10-attempt threshold, 10-minute lockout and observation window.

ENDPOINT VALIDATION

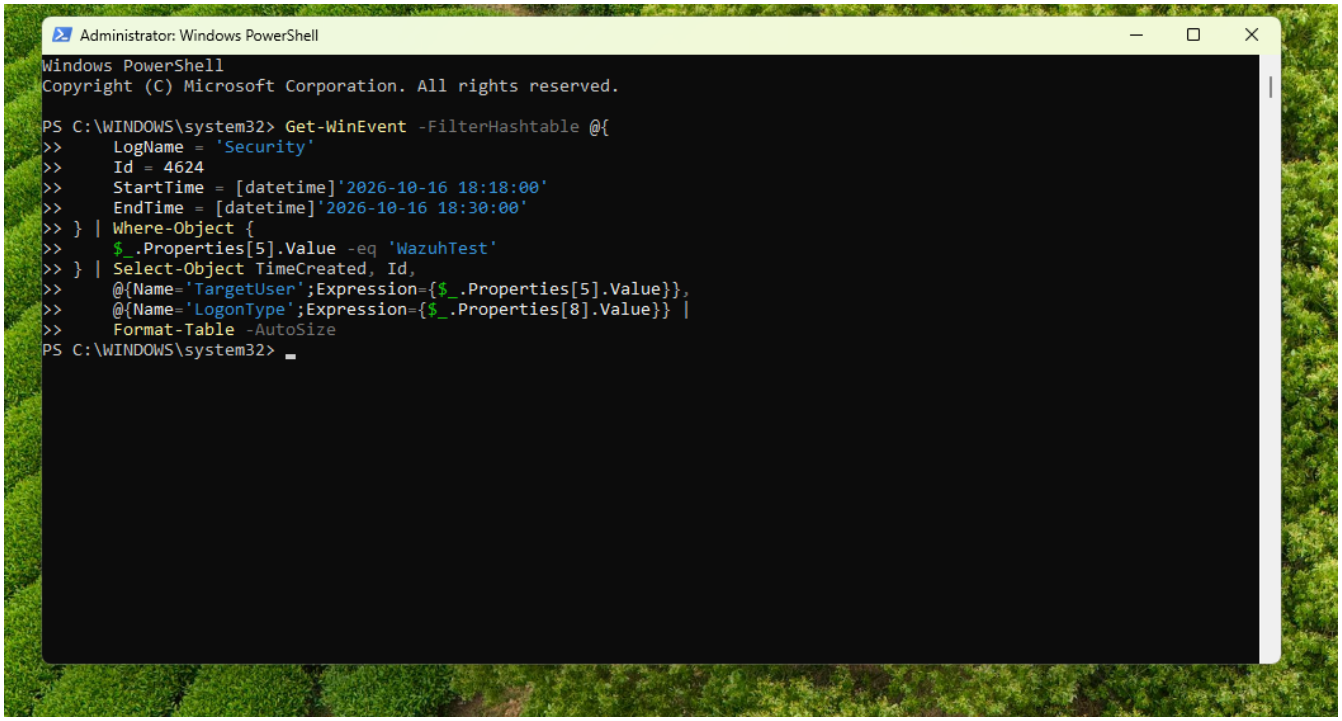


Exhibit D - Endpoint PowerShell: no matching successful logons (4624) in the reviewed window.

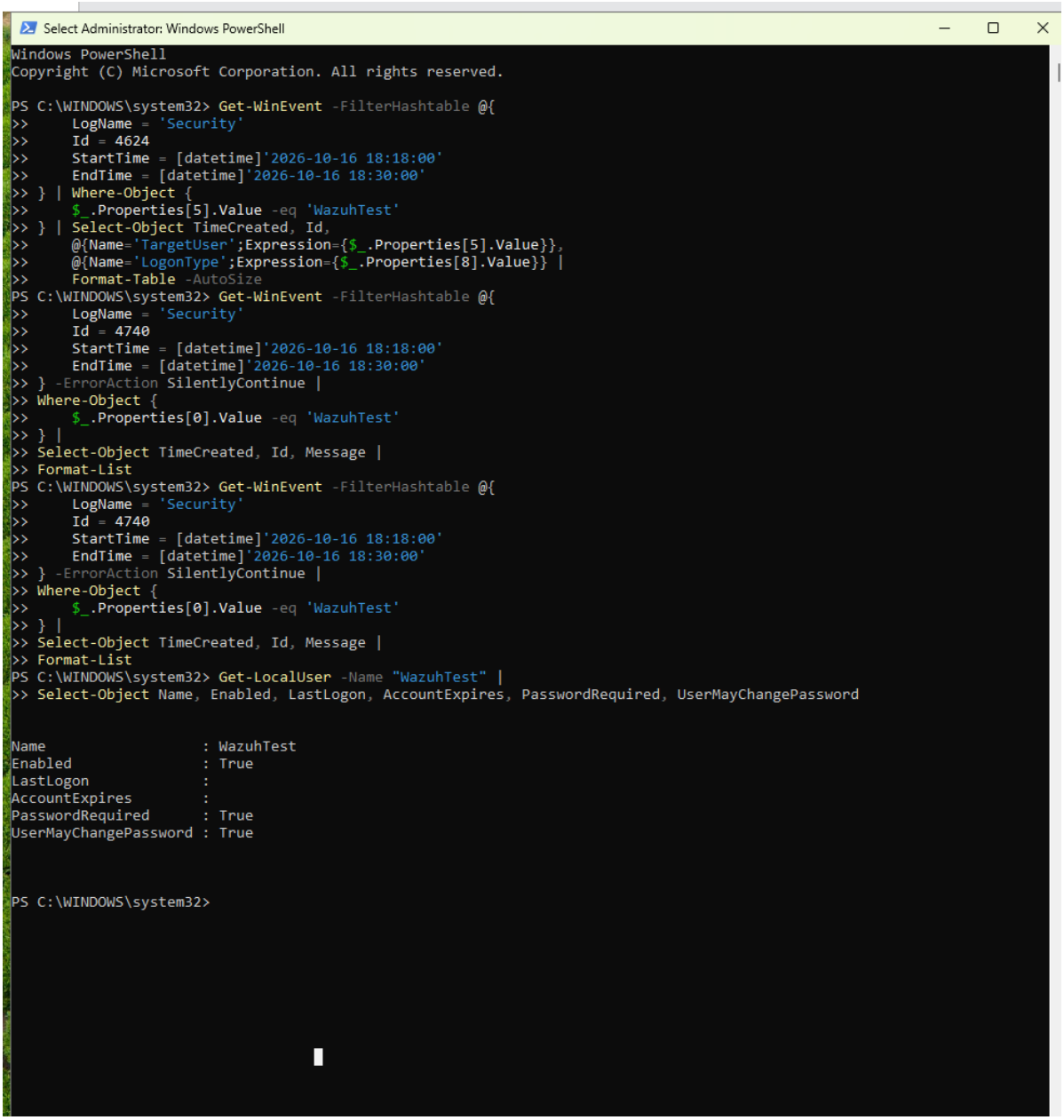


Exhibit E - Endpoint checks for 4740 lockouts and WazuhTest local account status.